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Hyperventilation-Aided Recovery for Extra Repetitions on Bench Press and Leg Press

Akihiro Sakamoto,[1] Hisashi Naito,[1] and Chin Moi Chow[2]

[1]Faculty of Health and Sports Science, Juntendo University, Inzai, Chiba, Japan; and [2]Discipline of Exercise and Sport Science, Faculty of Health Sciences, The University of Sydney, Lidcombe, New South Wales, Australia

Abstract

Sakamoto, A, Naito, H, and Chow, CM. Hyperventilation-aided recovery for extra repetitions on bench press and leg press. *J Strength Cond Res* 34(5): 1274–1284, 2020—Hyperventilation (HV)-induced alkalosis, an ergogenic strategy, improved repeated pedaling sprint performance through enhanced H^+ removal. However, it did not confer beneficial effects on other forms of exercises. This study investigated the benefits of HV-aided recovery on lifting repetitions and joint velocity during resistance training involving multiple joints and both concentric and eccentric contractions. Eleven power-trained men (mean ± *SD* age: 22.5 ± 4.3 years, training experience: 8.3 ± 3.6 years) performed 6 sets each of bench press and leg press at 80% 1 repetition maximum. Each set was continued until failure, with a 5-minute recovery between sets. In protocol A, HV was implemented for 30 seconds before the first, third, and fifth sets of each exercise (HV-aided recovery), whereas spontaneous breathing continued throughout the recovery before the second, fourth, and sixth sets (control recovery). In protocol B, the order of the HV and control recoveries was reversed. For both protocols, reductions in repetitions (range: −4.7% to −22.5%) and velocity (range: −23.1% to −37.7%) were consistently observed after control recovery ($p < 0.05$), whereas HV-aided recovery resulted in increased repetitions (range: +21.3% to +55.7%) and velocity (range: +6.3% to +15.3%) ($p < 0.05$) or no reductions in these measures from the previous set. The total repetitions performed across 6 sets (protocols A and B combined) were greater after the HV-aided than control recovery ($p \le 0.001$) in bench press (44 ± 10 vs. 36 ± 10 reps, increased by 27.1 ± 24.1%) and leg press (64 ± 9 vs. 50 ± 15 reps, increased by 35.2 ± 29.5%). Hyperventilation-aided recovery may boost the effectiveness of resistance training through increased training volume and lifting velocity.

Key Words: hypocapnia, respiratory alkalosis, muscular performance, fatigue, joint angular velocity

Introduction

Competitive athletes apply greater training volume, force, and velocity for effective strength and power gains (17,31,35,40) to achieve a winning edge (32,35). For subsequent sustained high power output, an adequate recovery time is required between training sets. This time is necessary for metabolic acid clearance, so that training volume and force-velocity development are not compromised. Accordingly, athletes may benefit from ergogenic aids that enhance recovery. Ergogenic aids including sodium bicarbonate ingestion and voluntary hyperventilation (HV), both induce alkalosis, target the acidic milieu within the muscle environment through buffering mechanisms (2,20,36).

In general, physical training for strength and power gains requires a high demand for adenosine triphosphate hydrolysis that is largely met by an increased rate of anaerobic glycolysis. The ensuing release of excessive amount of hydrogen ions (H^+) may result in the impairment of the excitation-contraction coupling system (13,15,28,30) and diminution of energy supply from phosphocreatine (PCr) and anaerobic glycolysis (23,25,34,42). These physiological changes are detrimental to muscular performance (fatigue) and may limit the training volume or velocity of muscular contraction. Sodium bicarbonate ingestion that induces metabolic alkalosis has been shown to enhance performance in intense whole-body intermittent exercise such as repeated sprints (2,3,20), although some studies observed no improvements (14,19). Sodium bicarbonate ingestion establishes a greater transmembrane H^+ gradient by means of H^+ uptake in the extracellular environment, thereby facilitating the efflux of H^+ from muscles (3,14,33). However, the evidence for its ergogenic effects for resistance training is limited and unconvincing (5,33,45).

Voluntary HV represents an alternative strategy for performance enhancement. It confers similar ergogenic effects without the adverse gastrointestinal effects that are often observed with sodium bicarbonate ingestion (26,45). The underlying buffering mechanism of HV differs from that of bicarbonate-induced metabolic alkalosis in that HV increases pH of both intracellular and extracellular environments through CO_2 excretion in excess of metabolic production yielding hypocapnia and respiratory alkalosis (7,8,12,21,27,44). The use of HV to aid recovery during repeated sets of maximal exercise has been studied recently with either positive or minor ergogenic effects (36,37).

Sakamoto et al. (36) demonstrated that, compared with the control breathing condition, greater power outputs were recorded in later sets of repeated pedaling sprints (10 seconds × 10 sets) when each sprint was preceded by a 30-second HV, implemented over the second half of 60-second interset recovery. Notably, HV can be initiated or terminated at will, without exogenous agents, thereby providing a greater utility in practice compared with metabolic alkalosis. Besides, the common symptoms of HV, such as dizziness, lightheadedness, or paresthesia (due to reduced cerebral and peripheral blood flow), were minimally or not felt by the subjects during the test (36).

Address correspondence to Dr. Akihiro Sakamoto, a-sakamo@juntendo.ac.jp.

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In advancing this research, Sakamoto et al. (37) investigated the effects of HV-aided recovery on torque outputs during repeated sets of maximal isokinetic knee extensions. The exercise consisted of 12 repetitions × 8 sets at $60°\cdot s^{-1}$ and 25 repetitions × 8 sets at $300°\cdot s^{-1}$, with a 40-second recovery between sets for both velocities. When HV was implemented in the last 30 seconds before the initiation of each exercise set, torque outputs at the beginning of the first and fourth sets at $60°\cdot s^{-1}$ were greater than those of control breathing condition. However, torque outputs of the other sets at $60°\cdot s^{-1}$ and all sets at $300°\cdot s^{-1}$ were not significantly different between the HV and control conditions. The study suggested that a single-joint exercise with concentric-only contractions may not have produced a sufficiently disturbed acid-base milieu for intervention, as evidenced by blood pH that was above 7.35 after the exercise. Therefore, it was suggested that resistance training involving multiple joints or muscle groups and both concentric and eccentric contractions be a future focus for the investigation of ergogenic potential of HV.

Accordingly, this study investigated the role of HV-aided recovery on lifting performance during repeated sets of strength-type bench press and leg press. Lifting performance was evaluated with the number of repetitions and the lifting velocity achieved in each training set preceded by either HV (HV-aided recovery) or normal spontaneous breathing (control recovery). This study tests the hypothesis that the HV-aided recovery yields a higher lifting repetitions and velocity compared with that after the control recovery. During strength-type resistance training, however, a 3- to 5-minute duration is implemented for the interset recovery to accomplish successful lifting over repeated sets (9).

Methods

Experimental Approach to the Problem

This study investigated the benefits of HV-aided recovery for lifting repetitions and velocity during 6 sets of bench press and 6 sets of leg press, where each training set was performed at 80% 1 repetition maximum (1RM) until lifting failure with a 5-minute recovery between sets. For the HV-aided recovery, subjects initially recovered with spontaneous breathing for 4 minutes and 30 seconds (from 0′00 to 4′30 of recovery), followed by 30-second of voluntary HV for the remaining 30 seconds (from 4′30 to 5′00) with the aim of lowering end-tidal partial pressure of CO_2 ($P_{ET}CO_2$) to 15–25 mm Hg (resting level 35–40 mm Hg) (36,37). For the control recovery, they breathed spontaneously for the entire 5-minute duration. To confirm the immediate and short-lasting effect of HV, the HV and control recoveries were allocated alternately across 6 sets. Where the HV-aided recovery was applied before the first, third, and fifth sets, as well as during additional recovery (explained below), was designated as protocol A, whereas in protocol B, the HV-aided recovery was applied before the second, fourth, and sixth sets (Figure 1). The other sets were preceded by the control recovery. Subjects underwent the exercise testing twice using both protocols A and B on separate occasions (with 48–72 hours apart). Wireless electrogoniometers attached at the elbow and the knee joints quantified the successful number of repetitions for each set and the lifting velocities during the eccentric and concentric phases for each repetition. Expired gas was monitored breath-by-breath during the exercise tests for respiratory rate (RR), expired tidal volume (TV_E), minute ventilation (V_E), and $P_{ET}CO_2$ to control for the methodological HV and to document the differences in the respiratory parameters between the HV and control recoveries. Blood pH, P_{CO_2}, and $[La^-]$ were also analyzed to report the changes in blood gas status as a result of HV and to estimate physiological strains imposed by the exercises.

Subjects

Eleven power-trained men with 7 throwers in athletics, 2 rugby players, 1 judo player, and 1 middle-distance sprinter participated in this study. Their mean ± *SD*: age, height, and body mass were 22.5 ± 4.3 years (range: 19–31 years), 176.1 ± 4.9 cm, and 90.0 ± 17.1 kg, respectively. Their 1RM strength was 129.1 ± 19.2 kg (range: 102.5–162.5 kg) for bench press and 208.1 ± 28.9 (range: 170.25–246.75 kg) for leg press. They had competed in their sports for 8.3 ± 3.6 years (with 5.4 ± 0.5 training days per week) and engaged in resistance training for 6.8 ± 3.1 years (3.9 ± 1.3 days per week) at the time of the experiment.

All subjects visited the laboratory on 4 separate occasions. During visit 1, they read the subject information statement and received verbal instructions regarding the experimental procedure and any associated risks and gave their written consent. Subjects were informed of the general physiological effects of HV as part of safety briefing. They were not told of the expected study outcomes. Visit 2 was a familiarization day. The subjects

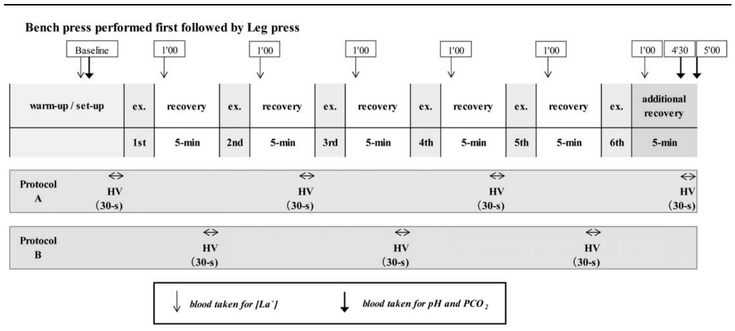

Figure 1. An overview of the experimental flow. Hyperventilation (HV) was implemented for 30 seconds from 4′30 to 5′00 of recovery before the first, third, and fifth exercise sets as well as during additional recovery in protocol A and before the second, fourth, and sixth sets in protocol B. Blood $[La^-]$ was measured before exercise (baseline) and then 1 minute after every exercise set (1′00 of recovery). Blood pH and P_{CO_2} were measured before exercise (baseline) and immediately before 4′30 and at 5′00 of additional recovery.

underwent 1RM testing for bench press and leg press (see below). Then, they were familiarized with the HV method as well as bench press and leg press procedures. This familiarization session served to confirm the subjects' capabilities to complete all the exercise protocols that were exhaustive, as well as providing practice in repetitive eccentric contractions affording a stress-evoked repeated-bout effect that is protective and adaptive in nature (24). The experimental exercise protocols were performed during visit 3 and visit 4 (protocol days). The familiarization day (visit 2) and the 2 protocol days (visits 3 and 4) were separated by 48–72 hours to avoid fatiguing effects. This duration was sufficient for the well-trained subjects to recover from major symptoms of eccentric muscle damage such as delayed-onset muscle soreness. Subjects were instructed to maintain their normal dietary intakes during the experimental period, with no intake of caffeinated or alcoholic drinks. To avoid hypoglycemia, they were asked to consume a normal Japanese meal (typically rich in carbohydrates, e.g., rice, noodles, and bread) 2 hours before each exercise test and drank ad libitum to maintain euhydration. All measurements were conducted in an air-conditioned training center (24–25° C). This study was approved by the Human Ethics Committee of Juntendo University, Graduate School of Health and Sports Science.

Procedures

Determinations of One Repetition Maximum Strength. Before commencing each exercise, subjects warmed up by performing 4–5 sets of each exercise of bench press and leg press with a progressive increase in load to their predicted 80–90% 1RM (predicted from their routine training). The loading weights and repetitions performed were freely chosen by the experienced subjects. Then, a 1RM weight was determined to the nearest 2.5 kg within 5 trials for each exercise, with a 3- to 5-minute recovery between trials. A successful trial was defined as a complete eccentric-concentric cycle across a full range of motion.

Familiarization. Each subject wore a face-fitting mask with expired gases monitored breath-by-breath using an aeromonitor (described below). Subjects were encouraged to breathe at an increased RR and TV to arrive at a preferred HV level without incurring a discomfort sensation, while achieving a reduction in $P_{ET}CO_2$ to 15–25 mm Hg (moderate hypocapnia). The duration of HV was set for 30 seconds based on our previous study, where a duration shorter or longer than 30 seconds did not improve intermittent sprint performance (38). Importantly, this duration of 30 seconds (with $P_{ET}CO_2$ range 15–25 mm Hg) is associated with minor adverse effects of HV such as dizziness and paresthesia, while effectively increasing blood pH (36,37). Then, the subjects undertook a full practice run of the experimental trial with 6 sets of bench press, followed by 6 sets of leg press. Before the practice run, subjects warmed up using the same method as for the 1RM testing (see above), except that the final warm-up loading was 80% of the actual 1RM determined. During the practice run, each exercise set was performed at 80% 1RM and continued until failure with a 5-minute recovery period between sets. Hyperventilation and control recoveries were implemented alternately across 6 sets to simulate either protocol A or B (see below for details). Respiratory rate or TV (or both) was adjusted when necessary according to the $P_{ET}CO_2$ levels. For example, when $P_{ET}CO_2$ went above the target range, subjects were asked to breathe faster or deeper and vice versa. The undertaken HV method and warm-up procedure were replicated on the actual protocol days.

Exercise Protocols. Subjects underwent 6 sets of bench press and then 6 sets of leg press at 80% 1RM with interset recovery of 5 minutes for both exercises. Hyperventilation was implemented for the last 30 seconds of recovery (from 4′30 to 5′00) before the start of first, third, and fifth sets (HV-aided recovery); alternating with control recovery before the other sets (spontaneous breathing for the entire 5-minute recovery). This protocol was designated as protocol A, whereas the reverse breathing implementation was designated as protocol B (Figure 1). The 2 protocols were allocated in a counterbalanced order among subjects. The subjects underwent both protocols A and B on separate days that were 48–72 hours apart.

Each exercise set was continued until lifting failure, with the successful number of repetitions and angular velocities of the elbow joint (bench press) or the knee joint (leg press) recorded. To achieve true lifting failure, for the familiarization day and the protocol days, bench press was performed on a Smith machine (F2 Free Weight Smith Machine; Nautilus Inc., Vancouver, WA) and leg press on a weight-stack machine (VR2, 4605 Seated Leg Press; Cybex International, Inc., Medway, MA), which secured the displacements of the bar and the footplate. Lifting was considered to have failed when unable to fully extend the elbow (bench press) or the knee joint (leg press) during the concentric phase. The lifting speed was self-selected, aiming to complete as many repetitions as possible before fatigue. Subjects were allowed to pause the lifting for no more than 3 seconds for breathing between repetitions (at the concentric-to-eccentric phase transition), especially near lifting failure.

Joint Angular Velocity and Repetition Measurements. Electrogoniometers (DTS 2D Electrical Goniometer, EM-852; Noraxon U.S.A, Inc., Scottsdale, AZ) were attached at the elbow and at the knee to calculate the mean joint angular velocities during the eccentric and concentric phases of every repetition. For the elbow joint, the goniometer was aligned with the humeral line (between the acromion process and the radial head) and the radial line (between the radial head and the styloid process of the radius). For the knee joint, the goniometer was aligned with the femoral line (between the greater trochanter and the lateral epicondyle of the femur) and the fibular line (between the lateral epicondyle of the femur and the lateral malleolus).

The signals were first collected using a Telemyo 2400R G2 telemetry system (Noraxon U.S.A., Inc.) and then digitalized by Powerlab (1,000 Hz) and processed using Powerlab Chart 8 software (ADInstruments, Castle Hill, Australia). Raw data were calibrated using 2 known angles and then low-pass filtered to reduce signal artifacts (10 Hz, bidirectional Butterworth filter). The angle data were then converted into angular velocity by the software using the central difference method (9 points window width). Using the angle and velocity data, the eccentric and concentric phases were identified for each repetition, and the mean angular velocities during the concentric and eccentric phases calculated. The goniometer data were also used to verify the number of successful repetitions per exercise set.

The numbers of repetitions achieved were variable among the exercise sets and among the subjects. The minimum number observed in this study was 2 repetitions; hence, the average of the first and second fastest velocities within each exercise set represented the joint angular velocity for that set (concentric and eccentric phases separated), which was then used for statistical analyses.

Respiratory Data. Respiratory rate, TV_E, V_E, and $P_{ET}CO_2$ were monitored breath-by-breath by an aeromonitor (AE100i; Minato Medical Science, Osaka, Japan). These respiratory parameters were displayed on a PC screen throughout the experiment to provide immediate feedback to guide the subjects' own target $P_{ET}CO_2$ value (previously determined in the familiarization session). Subjects were asked to inform the investigator if they experienced any discomfort sensations during the HV process. Before each test, the airflow sensor was calibrated using a 2-L syringe and O_2 and CO_2 sensors with known O_2 and CO_2 gas concentrations in accordance with the manufacturer's instructions. To report the differences in the respiratory profiles between the HV and control recoveries, the values recorded over the last 30 seconds of each recovery (from 4′30 to 5′00) were averaged. Only $P_{ET}CO_2$ values were averaged over the last 10 seconds of recovery (4′50–5′00) due to a slower rate of change to the target value range (15–25 mm Hg for the HV recovery). Figure 2 presents the mean ($\pm SD$) values of the respiratory parameters during the HV and control recoveries. The average RR and TV_E ($n = 11$) used for each HV intervention were, respectively, 46–50 breath·min^{-1} (range: 35.4–63.7 breath·min^{-1}) and 2,175–2,337 ml (range: 1,550–3,514 ml), achieving a fall in $P_{ET}CO_2$ to 16.3–23.4 mm Hg (range: 13.4–28.1 mm Hg). The average $P_{ET}CO_2$ after the control recovery was 35.0–40.9 mm Hg (range: 28.6–45.6 mm Hg) (Figure 2G and H).

Blood Gases and Lactate Measurements. After the last exercise set (sixth set), an additional 5-minute recovery was allocated for blood gas measurements (pH and P_{CO_2}) so as not to disrupt the protocol run. During this additional recovery, the HV-aided recovery was applied (HV from 4′30 to 5′00) for protocol A, whereas the control recovery was applied for protocol B (Figure 1). Whole capillary blood from the earlobe (approximately 60 μL) was collected into a heparinized capillary tube before the exercise (baseline) and then twice during the additional recovery, immediately before 4′30 and at 5′00 (Figure 1). The collected blood samples were analyzed using Cobas b221 System (Roche Diagnostics, Tokyo, Japan). The blood gas data served to examine the physiological strains of the exercises and the alterations of pH and P_{CO_2} in 30 seconds with or without HV. The measurements of blood pH and P_{CO_2} were occasionally unsuccessful due to device errors.

Blood was also sampled (20 μL) before exercise (baseline) and then at 1′00 of each 5-minute recovery (Figure 1) to identify the concentration of blood lactate ([La$^-$]) using a lactate analyzer (Biosen S-Line; EFK Diagnostics, Barleben, Germany). The [La$^-$] values served to estimate the levels of glycolytic challenges incurred during the exercise.

Statistical Analyses

To detect the main effect of exercise set, 1-way repeated-measures analysis of variances were performed on the number of repetitions and the joint angular velocity within each protocol (A or B) and within each exercise (bench press or leg press). When the assumption of sphericity was violated, significance was corrected using the Greenhouse-Geisser adjustment. Where the main effect of set was found significant, then pairwise comparisons with least significant difference (LSD) method were applied to examine the changes in these performance measures from the previous set.

The numbers of repetitions recorded after all the HV-aided recoveries or the control recoveries were, respectively, added up across the 2 protocols (3 sets per protocol × 2 protocols), presenting an aggregate number for each exercise. A paired *t*-test was used to detect a statistical difference in the total number of repetitions (HV vs. control recoveries) per exercise, with the effect size calculated using the Cohen's *d* formula. A paired *t*-test was also performed on the maximum blood [La$^-$] recorded to examine which exercise incurred a greater glycolytic challenge (bench press vs. leg press).

Linear mixed-model analyses (repeated measures) were used for blood pH and P_{CO_2} with occasional missing data points. When significant time effect was detected, pairwise comparisons (LSD method) were performed between time points of baseline vs. at 4′30 of additional recovery vs. at 5′00 of additional recovery for each protocol and exercise.

Statistical tests were performed using SPSS version 17.0 software (SPSS Inc., Chicago, IL). All values are expressed as mean ± *SD*. *p* values of equal to or less than 0.05 were considered statistically significant.

Results

No subjects reported discomfort sensations related to HV (dizziness, paresthesia, or discomfort in the chest) during the experiment.

The Number of Repetitions

From Figure 3A and B, significant set effects were found for both exercises and both protocols, with repetitions gradually decreasing ($p < 0.001$). Pairwise comparisons revealed that the control recovery consistently resulted in decreased repetitions by between −23.1% and −37.7% ($p < 0.01$). By contrast, for the sets that ensued the HV-aided recovery, the repetitions during bench press were maintained (no reductions), and during leg press, they were maintained or increased by between +21.3% and +55.7% ($p < 0.05$) from the previous set regardless of the protocols.

Figure 3C shows the total number of repetitions (protocols A and B) achieved after the HV-aided and control recoveries for each exercise. The total repetitions were significantly greater for the HV-aided recovery than for the control recovery in bench press (44 ± 10 vs. 36 ± 10 reps, increased by 27.1 ± 24.1%, $p = 0.001$) and leg press (64 ± 9 vs. 50 ± 15 reps, increased by 35.2 ± 29.5%, $p < 0.001$).

Mean Joint Angular Velocity

Figure 4 shows the mean joint angular velocities recorded during the eccentric and concentric phases (analyzed separately) for bench press (Figure 4A–D) and for leg press (Figure 4E–H). The main effect of set was significant ($p \leq 0.001$) for both exercises, both phases (concentric and eccentric), and both protocols (A and B). Similar to the findings for the repetitions, the joint velocity after the control recovery consistently showed a decline from the previous set by −4.7% to −22.5% ($p < 0.05$), except for the eccentric velocity of the sixth set of leg press in protocol A (Figure 4E), and the concentric velocity of the third set of leg press in protocol B (Figure 4H). By contrast, the HV-aided recovery was effective in preventing or reversing (range +6.3% to +15.3%, $p < 0.05$) the decline in the joint velocity from the previous set. The reversal of the decline in velocity, resulting from the HV-aided recovery, was more evident during leg press (Figure 4E–H) than bench press (Figure 4A–D).

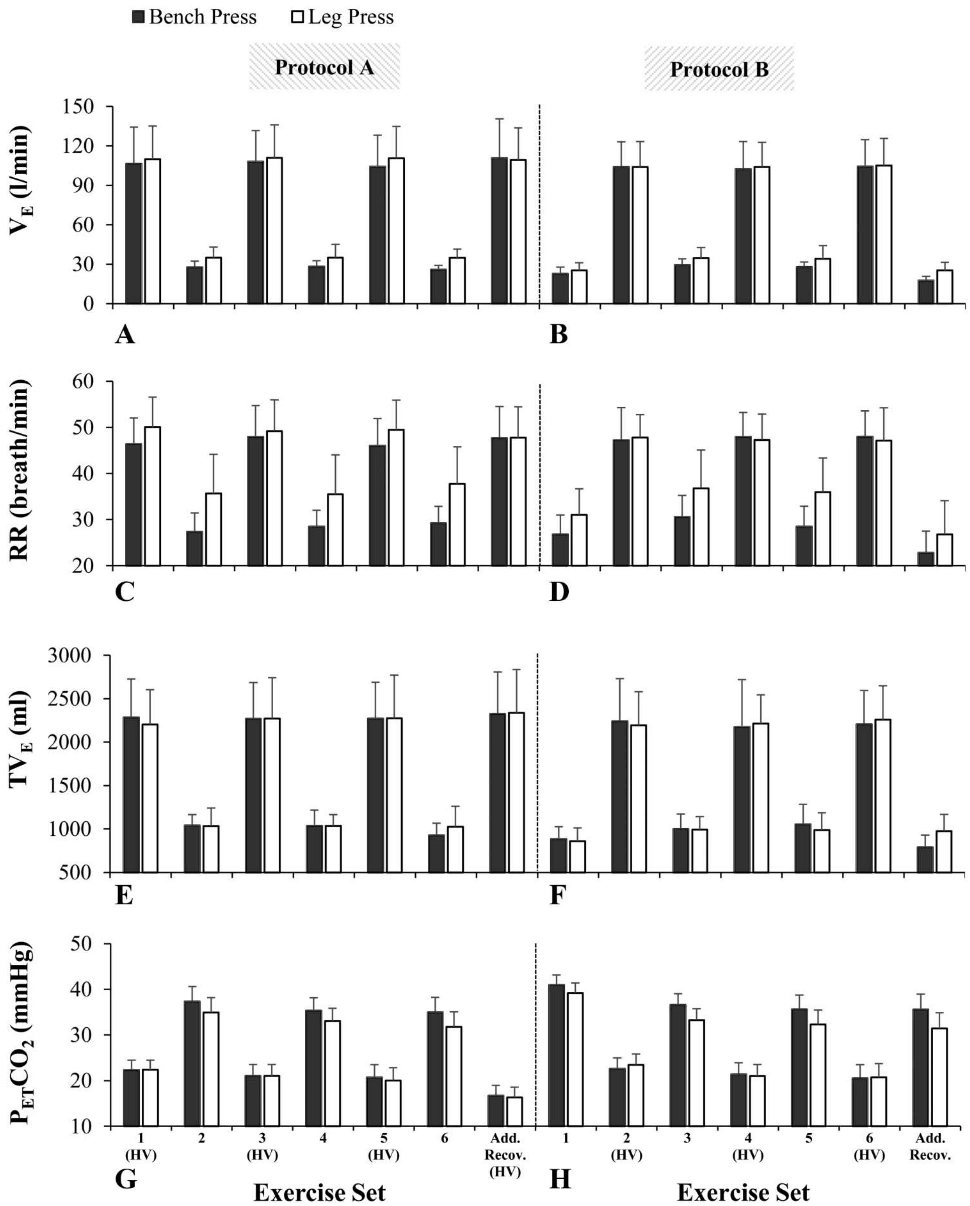

Figure 2. Comparisons of minute ventilation (V_E), respiratory rate (RR), expired tidal volume (TV_E), and end-tidal partial pressure of CO_2 ($P_{ET}CO_2$) with or without hyperventilation (mean ± *SD*, $n = 11$). Hyperventilation (HV) was implemented for 30 seconds from 4′30 to 5′00 of recovery immediately before the first, third, and fifth exercise sets as well as during additional recovery (Add. Recov.) in protocol A (A, C, E, and G) and before the second, fourth, and sixth exercise sets in protocol B (B, D, F, and H). The other sets were preceded by the control recovery. The values are the average over the last 30 seconds of each recovery, except for $P_{ET}CO_2$ that shows the average over the last 10 seconds of each recovery.

Blood pH and P_{CO_2}

Blood pH and P_{CO_2} changed significantly over time for both exercises and protocols ($p < 0.001$, Figure 5). Figure 5 illustrates that (a) after the exercise, blood pH and P_{CO_2} decreased from the baseline to 7.29–7.34 and 31.8–35.7 mm Hg, respectively, (b) the HV-aided recovery was effective in reversing the fall of pH to 7.39–7.42, while lowering the P_{CO_2} level further to 24.3–26.2 mm Hg, indicative of moderate hypocapnia (see 4′30 of additional recovery vs. 5′00 of additional recovery, Figure 5A and C), and (c) in the last 30 seconds of the control recovery, blood pH and P_{CO_2} did not significantly change but remained near 7.28–7.33 and 32.7–37.0 mm Hg (Figure 5B and D).

Blood [La⁻]

The maximum [La⁻] recorded during bench press and leg press was, respectively, 7.07 ± 1.51 vs. 9.82 ± 2.66 mM for protocol A and 6.68 ± 1.48 vs. 9.64 ± 2.87 mM for protocol B. Leg press incurred significantly greater [La⁻] values than bench press for both protocols ($p = 0.001$).

Discussion

This study corroborated the benefits of HV-aided recovery for lifting performance during heavy-loaded resistance training. For both protocols A and B, HV consistently prevented or reversed

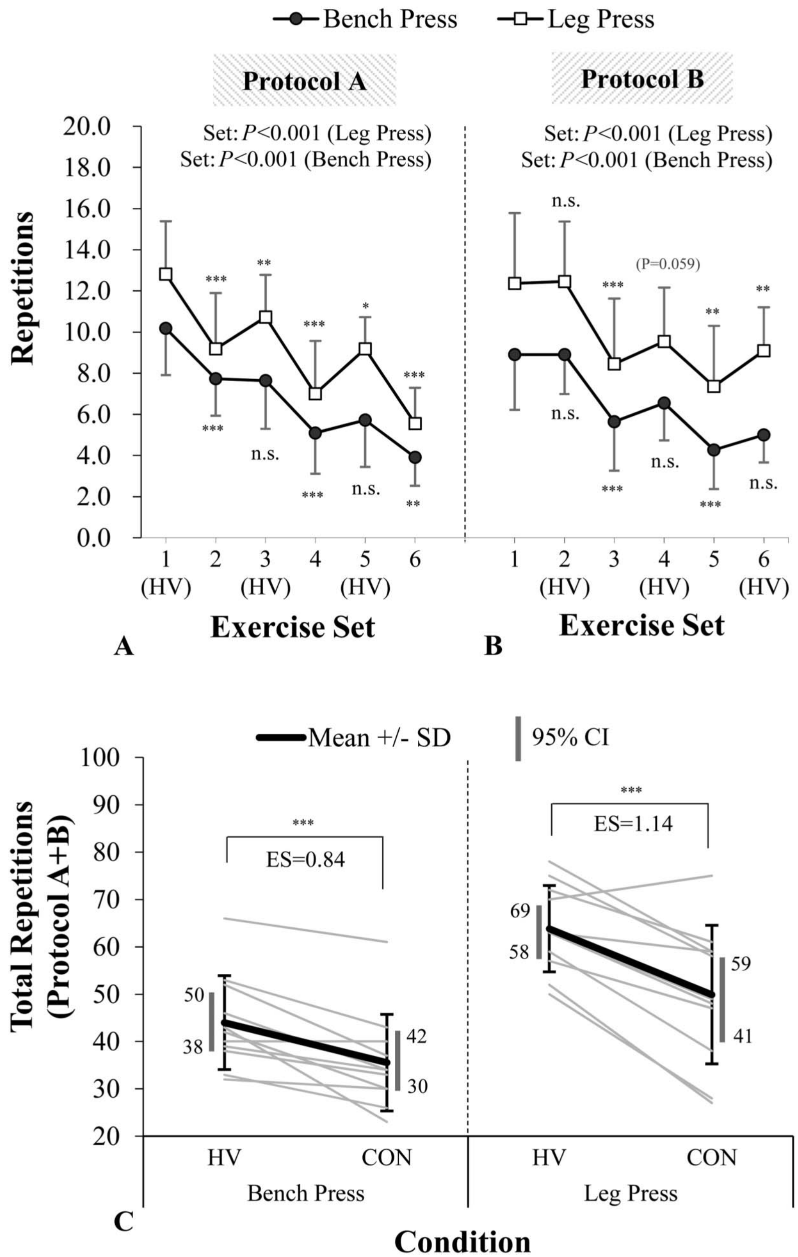

Figure 3. The number of repetitions (mean ± *SD*, *n* = 11) achieved in each exercise set (first–sixth) during bench press and leg press with the hyperventilation or control recovery. Hyperventilation (HV) was implemented for 30 seconds from 4′30 to 5′00 of recovery immediately before the first, third, and fifth exercise sets in protocol A (A) and before the second, fourth, and sixth sets in protocol B (B). The total numbers of repetitions (protocol A + B) from 6 exercise sets are shown in (C), comparing the control (CON) and hyperventilation (HV) recoveries. Bold lines in (C) represent the means of all subjects, whereas pale lines show individual data. ES, effect size determined by Cohen's *d*. 95% CI, 95% confidence interval (gray vertical bars). Significance levels for the main effect of set (1-way repeated-measures analysis of variance) are shown in the figures (A and B). Significant differences according to pairwise comparisons (vs. previous set) are indicated by asterisk, where ***$p \leq 0.001$, **$p \leq 0.01$, *$p \leq 0.05$, and n.s., nonsignificant.

the reduction in the number of repetitions and lifting velocities in the exercise set that followed, whereas these performance measures nearly always declined after the control recovery (Figures 3 and 4). Thus, the total repetitions attained across 6 exercise sets for the HV-aided recovery were greater by 9 ± 6 repetitions (an increase of 27.1 ± 24.1%) for bench press and by 14 ± 9 repetitions (an increase of 35.2 ± 29.5%) for leg press compared with the control recovery (Figure 3C). These findings clearly illustrate a positive ergogenic property of HV, rather than a chance occurrence.

Blood data, collected in the additional recovery period after the sixth set, confirmed that the HV method applied effectively reversed the fall in blood pH (Figure 5A) as P_{CO_2} continued to drop for time points from 4′30 to 5′00 (additional recovery)

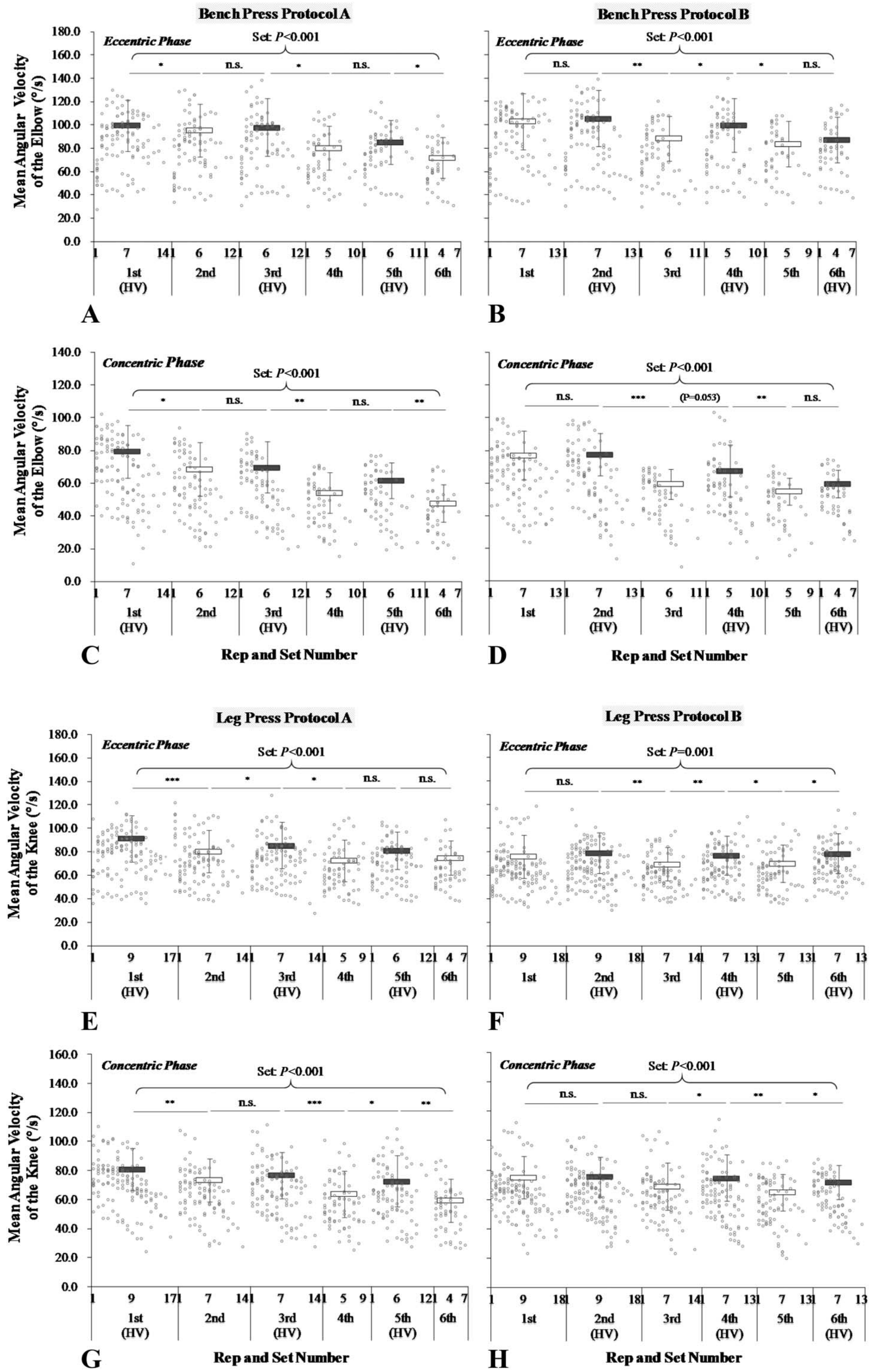

Figure 4. Mean joint angular velocities at the elbow joint during bench press for the eccentric phase (A) and the concentric phase (C) in protocol A and for the eccentric phase (B) and the concentric phase (D) in protocol B (mean ± *SD*, $n = 11$). Mean joint angular velocities at the knee joint during leg press for the eccentric phase (E) and the concentric phase (G) in protocol A and for the eccentric phase (F) and the concentric phase (H) in protocol B (mean ± *SD*, $n = 11$). Each rectangle represents the average of the first and second fastest mean angular velocities within the set. White boxes show the exercise sets preceded by the control recovery, and gray boxes show the exercise sets preceded by the hyperventilation recovery (HV). The pale circles show individual data for all repetitions. Significance levels for the main effect of set (1-way repeated-measures analysis of variance) are shown in the figure. Significant differences according to pairwise comparisons (vs. previous set) are indicated by asterisks, where ***$p \leq 0.001$, **$p \leq 0.01$, *$p \leq 0.05$, and n.s., nonsignificant.

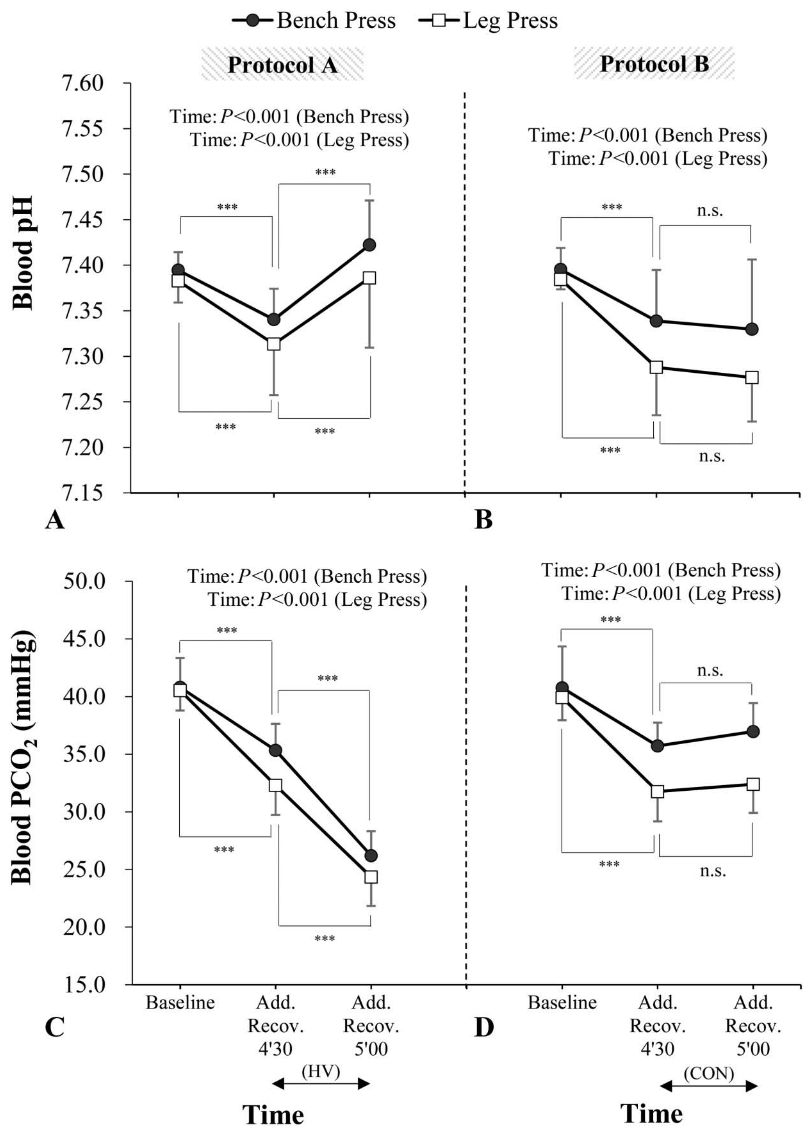

Figure 5. Changes in blood pH (A and B) and P_{CO_2} (C and D) between baseline vs. 4′30 of additional recovery (Add. Recov.) vs. 5′00 of additional recovery for bench press and leg press (mean ± *SD*, *n* = 11). Data at 4′30 and 5′00 of additional recovery in protocol A (A and C) show the changes in blood pH and P_{CO_2} before and after hyperventilation (HV), whereas those in protocol B (B and D) show the changes with control spontaneous breathing (CON). Significance levels for the main effect of set (1-way repeated-measures analysis of variance) are shown in the figure. Significant differences according to pairwise comparisons (vs. previous set) are indicated by asterisks, where ***$p \leq 0.001$ and n.s., nonsignificant.

(Figure 5C). During intense physical activities, excessive release of H^+ as a metabolic by-product is believed to be a cause of muscle fatigue through impaired excitation-contraction mechanisms (13,15,28,30) or diminution of energy supplies through PCr and anaerobic glycolysis (8,21,34,42). Hence, a faster restoration of the acid-base balance may facilitate the recovery of muscle's contractility. Previously, HV-induced respiratory alkalosis has been shown to sustain neuronal and muscle excitabilities (16,18,41) and provided indirect evidence for facilitated energy supplies from PCr and glycolysis (8,10,12). These mechanistic studies did not observe positive changes with performance outcomes, whereas in this study, HV effects were translated to improved strength recovery as reflected in the extra repetitions on bench press and leg press and improved joint angular velocity.

In this study, subjects used their preferred lifting velocities to achieve as many repetitions as possible before fatigue developed in each set. Naturally, they commenced each set of lifting at a velocity that was moderately fast, rather than using a steady velocity (Figure 4). Notably, HV improved the recovery of mean joint angular velocity during both eccentric and concentric phases (Figure 4). Sakamoto and Sinclair (39) reported that the number of repetitions was greater for faster lifting velocity at any given submaximal load, resulting from greater utilization of the properties of stretch-shortening cycle (SSC). The properties of SSC can be augmented with faster stretching and shortening speed of the involved muscle and shorter time elapsed at the phase transition (4,6,43). Faster stretching or eccentric speed in weight-loaded repetitive movements, however, necessitates a higher force to counter increased eccentric momentum at the phase transition, so that abrupt phase transition and fast muscle shortening are not compromised. Hence, the HV-aided recovery delivered 2 vital effects: (a) improved strength recovery and (b) sustained ability to exploit the SSC effects, both of which may have collectively contributed to the increased repetitions, indicative of improved muscular endurance.

To replicate the ergogenic effects, a best possible HV technique may need to be instituted. Sakamoto et al. (38) have recently reported that the ergogenic effects of HV may be impaired when the intervention duration is too short (15-second) or too long (45-second). Hyperventilation that is too short may not adequately impact on the acid-base restoration process for muscular performance during intense exercise. This also suggests that the application of HV may not be suitable for exercise programs with brief interset recoveries. When the intervention is too long, the detrimental effects of HV, such as decreased hemodynamics and slowed O_2 kinetics in both the periphery and the brain, may be exaggerated (7,11,12,29,44). In such a condition, O_2 and substrate delivery for PCr resynthesis and the Cori-cycle during the interset recovery may be impaired, leading to a performance decrement in the subsequent exercise set. The reduction of cerebral blood flow may also impair performance through reduced central motor commands (11,29). Furthermore, when discomfort sensation is felt during HV (i.e., discomfort in the chest due to forced ventilation and dizziness and paresthesia due to decreased hemodynamics and O_2 uptake), a prolonged intervention may aggravate these adverse symptoms, which may precipitate the development of central fatigue. To minimize these detrimental effects, the HV duration was set for 30 seconds in this study based on our previous studies on repeated pedaling sprints (36,38). Moreover, the subjects in this study were allowed to choose their preferred HV technique by manipulating both RR and TV_E to lower $P_{ET}CO_2$ to below 25 mm Hg. Consequently, no subjects experienced discomfort sensation during the experiment. In the previous HV studies, however, discomfort sensation was occasionally experienced when a constant RR was prescribed for all subjects (50 or 60 breath·min^{-1}), with only TV_E able to be manipulated (36–38). The attempt made in this study to overcome these previous concerns possibly reinforced the ergogenic potential of HV.

This study used multiple-joint exercises involving both eccentric and concentric contractions. The changes in the blood parameters were greater for leg press than bench press. The blood pH fell below 7.330 and [La$^-$] increased above 6.68 mM after bench press, whereas the corresponding values were below 7.277 and above 9.64 mM after leg press. These differences may be explained by the larger muscle mass involved in leg press. Accordingly, the ergogenic effects of HV were more pronounced for leg press than bench press (Figures 3 and 4). This is consistent with the previous notion that the ergogenic effects of induced alkalosis, of both metabolic and respiratory origins, manifest best under conditions of substantial anaerobic metabolic challenges or severe acidosis (2,5,14,27,33,44,45). In an earlier study by Sakamoto et al. (37), the ergogenic effects of HV were only minor when tested on repeated maximal isokinetic knee extensions (12 reps × 8 sets at 60°·s^{-1} and 25 reps × 8 sets at 300°·s^{-1} with a 40-second recovery between sets). Despite significant reduction in torque output, these single-joint concentric-only actions were insufficient to produce systemic metabolic disturbances; hence, fatigue may have been localized in the knee extensors only. The changes in blood pH and [La$^-$] after the exercise in this study (37) were indeed smaller compared with the present study.

To date, the evidence for the ergogenic effect of metabolic alkalosis through sodium bicarbonate ingestion was limited and unconvincing for resistance training (5,33,45). Metabolic alkalosis did not improve the number of repetitions performed during exhaustive heavy-loaded leg presses, where fatigue was induced by 5 sets of leg presses at 12RM or 70% 1RM, with a 90-second recovery between sets (33,45). The authors suggested that their resistance training tests were not as metabolically challenging as whole-body intermittent activities, so that the ergogenic properties of metabolic alkalosis were not expressed. A more recent study by Carr et al. (5) used multiple exercises of lower body (back squat, inclined leg press and knee extension), performing 4 sets per exercise at 10–12RM loading with a 60–90 seconds between sets (hypertrophy-type resistance training). The protocol of Carr et al. was aimed at intensifying metabolic stress compared with the earlier leg press studies. Consequently, the number of repetitions achieved was greater with metabolic alkalosis than the control condition when the repetitions of 2 or more exercises were combined. However, no significant condition effect was evident on repetitions when analyzed per exercise.

On the contrary, the present study provided evidence for increased repetitions through HV for each exercise (bench press and leg press) (Figure 3) despite a long interset recovery (5-minute). This recovery period was much longer than that used in the aforementioned metabolic alkalosis studies (60–90 seconds). However, the blood pH recorded at 4′30 after the last exercise set (Figure 5A and B), particularly after leg press, was similar to the end-exercise pH values recorded in the previous studies (pH = 7.25–7.28) (5,33,45). Remarkably, a sustained or increased lifting repetitions was already evident after the second or third exercise set in the present study (Figure 3A and B), where metabolic acidosis may not have been so severe. These observations propose that induced alkalosis of respiratory origin (i.e., HV) represents a more potent means to elicit ergogenic effects than metabolic alkalosis, although the mechanisms remain unclear. Importantly, the present experimental design revealed the ease of application of HV, which can be delivered and terminated on demand without persistent effects when stopped. Given its practical utility, HV is suggested as an alternative training strategy for countering fatigue, particularly when fatigue arises from severe acidic muscle milieu. However, it remains to be investigated whether additional training volumes and attenuated declines in velocity through HV-aided recovery will be translated to greater strength and power gains after long-term usage.

Voluntary HV may help to increase central motor command outflows through the mechanisms of the Setchenov phenomenon (1,22). A voluntary activation of nonfatigued muscles, i.e., the respiratory muscles during HV, may stimulate the facilitatory part of the reticular formation. Consequently, the feedback nerve impulses from fatigued muscles that cause an inhibition of the voluntary effort (central fatigue) may be countered. Compared with an exercise set attempted from a quiet rest, voluntary activations of the respiratory muscles may establish a more suited preparatory state with restored arousal levels. We suggest that this sort of invigorating effect may further explain the enhanced performance with HV. Furthermore, this mechanism may explain why HV improved lifting performance of resistance training more clearly than previous metabolic alkalosis aids. Yet, future research is warranted to ascertain that HV really triggers the Setchenov phenomenon.

In the current study, highly trained men were recruited and each exercise set performed at 80% 1RM until true lifting failure. The findings, therefore, may be limited to populations with skilled resistance training background or similar types of resistance training. The ergogenic effects of HV may not be assured when applying to other muscle groups, for different numbers of sets or recovery durations, or when each set is not continued until failure. Given the more pronounced improvements in lifting performance observed for leg press compared with bench press, it is likely that exercise types affect the effectiveness of HV. For a wider generalization, the effectiveness of HV needs to be investigated under many conceivable training situations.

Practical Applications

Highly resistance-trained individuals, who have reached their performance limits, may take advantage of performance enhancing aids to achieve higher repetitions and sustained high power output during their training. The resulting increase in training volume and speed of muscular contraction may offer further adaptive stimuli and motivation. Hyperventilation-aided recovery (46–50 breath·min^{-1}, 2.2–2.3 L/breath, 30-second duration, $P_{ET}CO_2$ = 15–25 mm Hg) represents an ergogenic strategy that has shown positive performance effects with minor side effects. It is a voluntary act and can be immediately commenced or terminated without persistent effects. Notably, the performance-enhancing effects are observed in highly specific intense intermittent exercise and athletes with skilled resistance training background. Accordingly, the current findings cannot be generalized to all training types and populations.

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